

VACA KEY, FL, USA

WMO: 997356

Lat: 24.711N

Lon: 81.107W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 54.9	(c) 57.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 25.9	(k) 67.3	(l) 23.6	(m) 65.7	(n) 11.9	(o) 20	(p) 0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.0	(c) 91.7	(d) N/A	(e) 90.3	(f) N/A	(g) 89.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.8	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.8	(b) 19.2	(c) 17.3	(e) 50.7	(f) 94.2	(g) 3.8	(h) 2.4	(i) 47.9	(j) 95.9	(k) 45.7	(l) 97.3	(m) 43.6	(n) 98.7	(o) 40.8	(p) 100.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.3	69.9	72.1	73.9	77.2	80.4	83.6	84.8	85.1	83.8	80.8	74.8	72.4
	DBStd	6.58	5.75	5.69	4.36	3.41	2.36	2.33	2.18	2.01	2.23	3.39	4.61	5.87
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	60	25	15	4	0	0	0	0	0	0	0	1	15
	CDD50	10318	617	620	742	816	941	1009	1077	1088	1013	955	745	695
	CDD65	4904	177	215	281	366	476	559	612	623	563	490	296	246
	CDH74	48004	541	890	1215	2550	4669	6879	8032	8319	7017	5070	1709	1113
	CDH80	16032	14	32	88	325	959	2701	3633	3919	2794	1326	178	63
(14) Wind	WSAvg	7.3	8.9	7.2	7.8	7.3	6.9	5.7	5.2	5.2	6.2	9.0	9.6	8.4
(15) Precipitation	PrecAvg	40.8	2.0	1.7	1.5	2.0	2.9	4.6	3.7	5.9	7.0	5.9	1.8	1.8
	PrecMax	55.4	8.2	6.1	3.5	6.1	9.1	14.7	8.3	19.3	15.2	17.2	6.7	5.4
	PrecMin	27.8	0.0	0.0	0.3	0.1	0.5	0.2	0.8	1.1	3.6	1.5	0.0	0.0
	PrecStd	7.4	2.0	1.6	1.0	1.5	2.4	3.3	1.9	3.3	2.4	4.1	1.8	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.4	82.4	84.8	88.7	88.9	91.9	93.3	93.3	92.5	90.4	85.4	83.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	79.9	81.0	82.4	84.9	86.8	89.9	91.4	91.4	90.4	87.4	83.6	82.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	78.5	79.9	80.5	83.0	85.3	88.6	89.8	89.8	88.6	86.1	82.2	80.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	77.2	78.5	79.1	81.8	84.1	87.4	88.4	88.7	87.3	84.9	81.0	79.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.2	6.8	6.8	6.5	7.2	7.5	8.0	7.5	6.1	6.0	6.5
		MCDBR	8.4	7.2	8.3	8.3	8.7	9.1	9.3	9.3	9.5	7.9	6.8	6.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.363	0.377	0.383	0.401	0.425	0.443	0.451	0.434	0.423	0.409	0.378	0.370
	taud		2.549	2.509	2.478	2.406	2.392	2.410	2.388	2.460	2.457	2.486	2.518	2.547
	Ebn at Noon		285	287	291	285	277	269	267	273	274	274	277	277
	Edh at Noon		29	32	35	38	39	38	38	36	35	33	30	28
(44) All-Sky Solar Radiation	RadAvg		1236	1528	1878	2119	2136	1980	2021	1921	1702	1530	1321	1165
	RadStd		91	101	77	94	188	152	96	77	81	106	65	100

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	+0.68	+1.57	N/A	N/A	N/A	N/A	N/A	-23	+209	+187

Nomenclature: See separate page